

CS 264 Lab Report

In this lab session, the task assigned was to design, simulate, and implement a digital stopwatch using the provided design resources. The given objectives included developing a debouncer circuit to ensure reliable push-button inputs, implementing a finite state machine to control the stopwatch operation, and designing the necessary counting logic for accurate time measurement. Additionally, a testbench was to be completed to provide appropriate stimulus for functional verification through simulation, followed by implementation and testing of the design on the hardware board.

Description:

This module implements a digital stopwatch using a finite state machine (FSM). It supports start, stop, and reset operations, generates a 1 Hz timing signal from the system clock, and displays elapsed time in seconds. Here is the relevant code -

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;

entity Stop_watch is
  Generic (
    CLK_FREQ_HZ : integer := 100_000_000; -- Default 100MHz
    DEBOUNCE_MS : integer := 20           -- Default 20ms
  );
  Port (
    timer_start : in  STD_LOGIC;
    timer_stop  : in  STD_LOGIC;
    timer_reset : in  STD_LOGIC;
    time_out    : out STD_LOGIC_VECTOR(7 downto 0);
    clk         : in  STD_LOGIC
  );
end Stop_watch;

architecture Behavioral of Stop_watch is

  component debounce_logic
    Generic (
      CLK_FREQ : integer;
      DEBOUNCE_MS : integer
    );
    Port (
      Key_press : in  STD_LOGIC;
      debounced_out : out STD_LOGIC;
      clk : in  STD_LOGIC
    );
  end component;
```

```
signal start_deb, stop_deb, reset_deb : STD_LOGIC;
```

```
type state_type is (IDLE, RUNNING, STOPPED);
```

```
signal current_state : state_type := IDLE;
```

```
-- Adjusted counter size logic based on Generic
```

```
signal counter_1hz : integer range 0 to CLK_FREQ_HZ := 0;
```

```
signal tick_1hz : STD_LOGIC := '0';
```

```
signal seconds : unsigned(7 downto 0) := (others => '0');
```

```
begin
```

```
-- Pass Generics to Debouncers
```

```
Debounce_Start: debounce_logic
```

```
generic map (CLK_FREQ => CLK_FREQ_HZ, DEBOUNCE_MS => DEBOUNCE_MS)
```

```
port map (Key_press => timer_start, debounced_out => start_deb, clk => clk);
```

```
Debounce_Stop: debounce_logic
```

```
generic map (CLK_FREQ => CLK_FREQ_HZ, DEBOUNCE_MS => DEBOUNCE_MS)
```

```
port map (Key_press => timer_stop, debounced_out => stop_deb, clk => clk);
```

```
Debounce_Reset: debounce_logic
```

```
generic map (CLK_FREQ => CLK_FREQ_HZ, DEBOUNCE_MS => DEBOUNCE_MS)
```

```
port map (Key_press => timer_reset, debounced_out => reset_deb, clk => clk);
```

```
-- 1 Hz Generator
```

```
process(clk)
```

```
begin
```

```
if rising_edge(clk) then
```

```
if counter_1hz = CLK_FREQ_HZ - 1 then
```

```
counter_1hz <= 0;
```

```
tick_1hz <= '1';
```

```
else
```

```
counter_1hz <= counter_1hz + 1;
```

```
tick_1hz <= '0';
```

```
end if;
```

```
end if;
```

```
end process;
```

```
-- FSM and Counter
```

```
process(clk)
```

```
begin
```

```
if rising_edge(clk) then
```

```
if reset_deb = '1' then
```

```
current_state <= IDLE;
```

```
seconds <= (others => '0');
```

```
else
```

```
case current_state is
```

```
when IDLE =>
```

```
if start_deb = '1' then
```

```

        current_state <= RUNNING;
    end if;

    when RUNNING =>
        if stop_deb = '1' then
            current_state <= STOPPED;
        end if;

        when STOPPED =>
            if start_deb = '1' then
                current_state <= RUNNING;
            end if;
        end case;
    end case;

    if current_state = RUNNING and tick_1hz = '1' then
        seconds <= seconds + 1; -- Will wrap automatically at 255
    end if;
end if;
end if;
end process;

time_out <= std_logic_vector(seconds);

end Behavioral;

```

Description:

This module implements debounce logic for mechanical push buttons. It filters out unwanted glitches caused by switch bouncing and provides a stable output signal for reliable control input detection. Here is the relevant code -

```

library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;

entity debounce_logic is
    Generic (
        CLK_FREQ    : integer := 100_000_000;
        DEBOUNCE_MS : integer := 20
    );
    Port (
        clk          : in  STD_LOGIC;
        Key_press    : in  STD_LOGIC;
        debounced_out : out STD_LOGIC
    );
end debounce_logic;

```

```

architecture Behavioral of debounce_logic is
    constant MAX_COUNT : integer := (CLK_FREQ / 1000) * DEBOUNCE_MS;
    signal counter      : integer range 0 to MAX_COUNT := 0;
    signal sync1, sync2 : std_logic := '0';
    signal button_stable : std_logic := '0';
    signal prev_stable   : std_logic := '0';

begin

    -- Edge detection
    debounced_out <= '1' when (button_stable = '1' and prev_stable = '0') else '0';

    process(clk)
    begin
        if rising_edge(clk) then
            -- 1. Synchronize Input
            sync1 <= Key_press;
            sync2 <= sync1;

            -- 2. Store previous stable state for edge detection
            prev_stable <= button_stable;

            -- 3. Debounce Logic
            if (button_stable /= sync2) then
                -- Input is different than our stable state
                if (counter < MAX_COUNT) then
                    counter <= counter + 1;
                else
                    -- We have waited long enough; commit the change
                    button_stable <= sync2;
                    counter <= 0;
                end if;
            else
                -- Input matches stable state; reset counter
                counter <= 0;
            end if;
        end if;
    end process;

end Behavioral;

```

Description:

This testbench verifies the functional correctness of the stopwatch design by simulating clock signals and control inputs such as start, stop, and reset, and observing the corresponding output behavior.

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;
```

```
entity tb_stop_watch is
end tb_stop_watch;
```

```
architecture Behavioral of tb_stop_watch is
```

```
    component Stop_watch
        Generic (
            CLK_FREQ_HZ : integer;
            DEBOUNCE_MS : integer
        );
        port (
            timer_start : in  STD_LOGIC;
            timer_stop  : in  STD_LOGIC;
            timer_reset  : in  STD_LOGIC;
            time_out    : out STD_LOGIC_VECTOR(7 downto 0);
            clk         : in  STD_LOGIC
        );
    end component;
```

```
    signal timer_start : STD_LOGIC := '0';
    signal timer_stop  : STD_LOGIC := '0';
    signal timer_reset  : STD_LOGIC := '0';
    signal time_out    : STD_LOGIC_VECTOR(7 downto 0);
    signal clk         : STD_LOGIC := '0';
```

```
    constant CLK_PERIOD : time := 10 ns;
```

```
begin
```

```
    uut: Stop_watch
        generic map (
            CLK_FREQ_HZ => 100, -- SIMULATION ONLY: 100 cycles = 1 second
            DEBOUNCE_MS => 1    -- SIMULATION ONLY: 1ms debounce
        )
        port map (
            timer_start => timer_start,
            timer_stop  => timer_stop,
            timer_reset  => timer_reset,
            time_out    => time_out,
            clk         => clk
        );
```

```
    clk_process: process
    begin
        clk <= '0';
        wait for CLK_PERIOD / 2;
        clk <= '1';
```

```

    wait for CLK_PERIOD / 2;
end process;

stim_proc: process
begin
    report "===== TEST START =====";

    -- Reset
    timer_reset <= '1';
    wait for 200 ns; -- Long enough for 1ms debounce? No, but reset logic handles it.
    timer_reset <= '0';
    wait for 100 ns;

    -- Test 1: Start counting
    -- NOTE: We must hold button longer than DEBOUNCE_MS (which we set to 1ms
above)
    -- 1ms = 1,000,000 ns. But since we scaled frequency to 100Hz,
    -- the debouncer math changes.
    -- Safest bet in sim: Hold button for 5 clock cycles.

    report "Test 1: Press START";
    timer_start <= '1';
    wait for 200 ns; -- Hold long enough for debouncer
    timer_start <= '0';

    -- Wait for 5 "seconds" (5 * 100 cycles * 10ns = 5000ns)
    wait for 5000 ns;

    -- Test 2: Stop
    report "Test 2: Press STOP";
    timer_stop <= '1';
    wait for 200 ns;
    timer_stop <= '0';

    wait for 2000 ns; -- Verify count doesn't change

    -- Test 3: Resume
    report "Test 3: Press START (Resume)";
    timer_start <= '1';
    wait for 200 ns;
    timer_start <= '0';
    wait for 3000 ns;

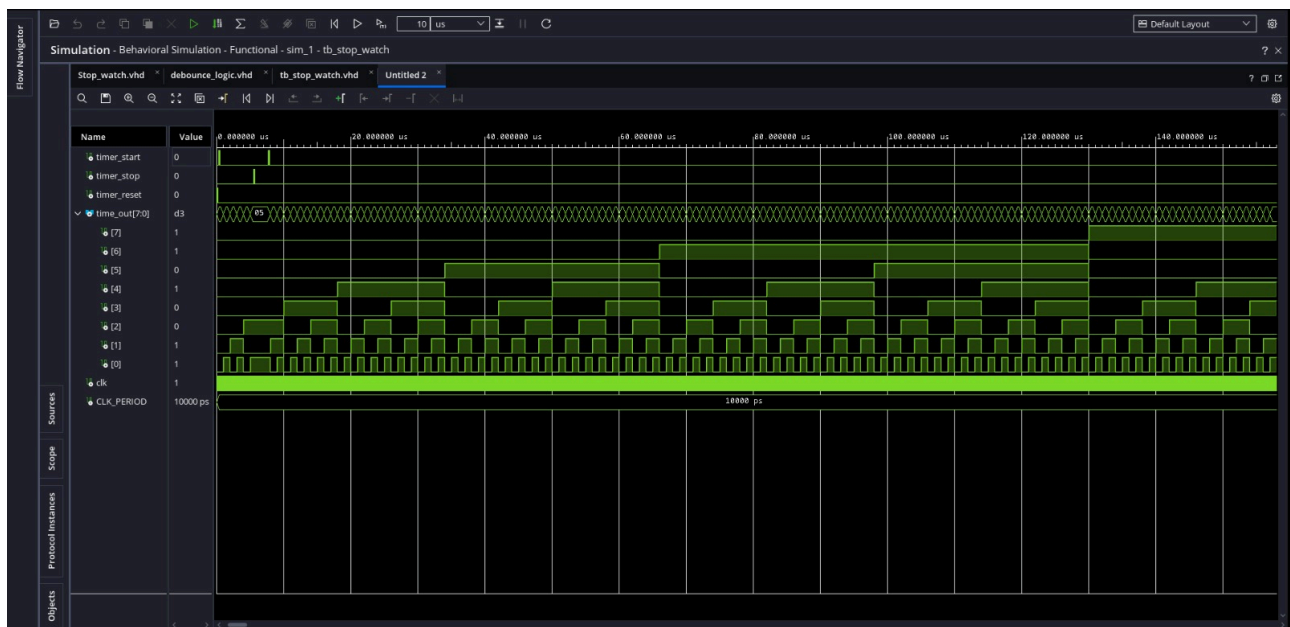
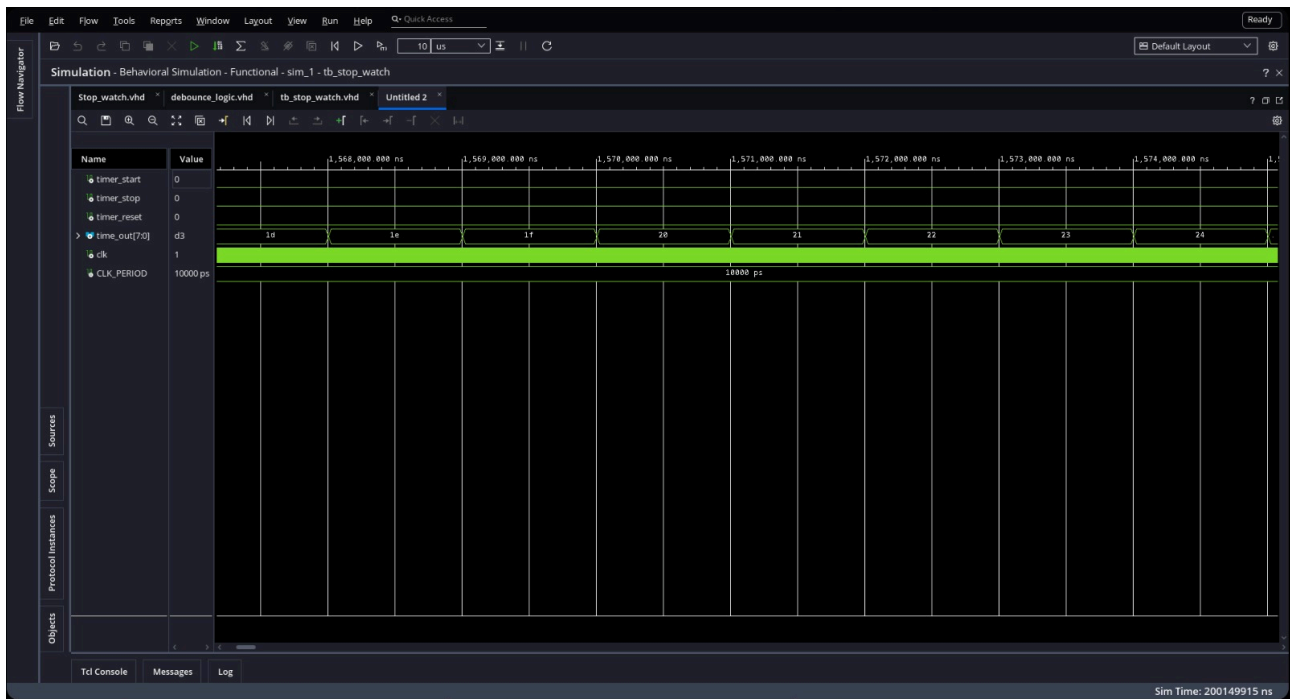
    report "===== TEST FINISHED =====";
    wait;
end process;
end Behavioral;

```

Simulation Results:

The stopwatch design was simulated to verify its functional correctness. The simulation waveform confirms correct state transitions, accurate time counting, and

proper response to start, stop, and reset control signals. Here are some simulation waveform snapshots -



Synthesis Results:

The design was synthesized successfully without errors. These synthesis logs confirms correct RTL interpretation, proper resource utilization, and compatibility of

the design with the target FPGA architecture.

WARNING: [filemgmt 56-199] Attempt to get parsing info during refresh. "On-the-fly" syntax checking information may be incorrect. [/home/user/Workspace/vivado/stopwatch/stopwatch.srcs/sim_1/new/tb_stop_watch.vhd:]

WARNING: [filemgmt 56-199] Attempt to get parsing info during refresh. "On-the-fly" syntax checking information may be incorrect. [/home/user/Workspace/vivado/stopwatch/stopwatch.srcs/sources_1/new/debounce_logic.vhd:]

WARNING: [filemgmt 56-199] Attempt to get parsing info during refresh. "On-the-fly" syntax checking information may be incorrect. [/home/user/Workspace/vivado/stopwatch/stopwatch.srcs/sources_1/new/Stop_watch.vhd:]

WARNING: [filemgmt 56-199] Attempt to get parsing info during refresh. "On-the-fly" syntax checking information may be incorrect. [/home/user/Workspace/vivado/stopwatch/stopwatch.srcs/sources_1/new/debounce_logic.vhd:]

WARNING: [filemgmt 56-199] Attempt to get parsing info during refresh. "On-the-fly" syntax checking information may be incorrect. [/home/user/Workspace/vivado/stopwatch/stopwatch.srcs/sources_1/new/debounce_logic.vhd:]

launch_simulation

Command: launch_simulation

INFO: [Vivado 12-25911] Executing simulation for 'XSim' simulator

INFO: [Project 1-5737] Running on host arch (64) - unknown

INFO: [Vivado 12-28535] Process started - Fri Jan 30 15:48:23 2026

WARNING: [filemgmt 56-199] Attempt to get parsing info during refresh. "On-the-fly" syntax checking information may be incorrect. [/home/user/Workspace/vivado/stopwatch/stopwatch.srcs/sim_1/new/tb_stop_watch.vhd:]

WARNING: [filemgmt 56-199] Attempt to get parsing info during refresh. "On-the-fly" syntax checking information may be incorrect. [/home/user/Workspace/vivado/stopwatch/stopwatch.srcs/sources_1/new/debounce_logic.vhd:]

WARNING: [filemgmt 56-199] Attempt to get parsing info during refresh. "On-the-fly" syntax checking information may be incorrect. [/home/user/Workspace/vivado/stopwatch/stopwatch.srcs/sources_1/new/Stop_watch.vhd:]

WARNING: [filemgmt 56-199] Attempt to get parsing info during refresh. "On-the-fly" syntax checking information may be incorrect. [/home/user/Workspace/vivado/stopwatch/stopwatch.srcs/sources_1/new/Stop_watch.vhd:]

WARNING: [filemgmt 56-199] Attempt to get parsing info during refresh. "On-the-fly" syntax checking information may be incorrect. [/home/user/Workspace/vivado/stopwatch/stopwatch.srcs/sources_1/new/Stop_watch.vhd:]

INFO: [Vivado 12-12493] Simulation top is 'tb_stop_watch'

INFO: [Vivado 12-5682] Launching behavioral simulation in '/home/user/Workspace/vivado/stopwatch/stopwatch.sim/sim_1/behav/xsim'

INFO: [SIM-utils-51] Simulation object is 'sim_1'

INFO: [SIM-utils-72] Using boost library from '/home/user/Vivado/2025.2/tps/boost_1_72_0'

INFO: [SIM-utils-54] Inspecting design source files for 'tb_stop_watch' in fileset 'sim_1'...

INFO: [USF-XSim-97] Finding global include files...

INFO: [USF-XSim-98] Fetching design files from 'sim_1'...

INFO: [USF-XSim-2] XSim::Compile design

INFO: [USF-XSim-61] Executing 'COMPILE and ANALYZE' step in '/home/user/Workspace/vivado/stopwatch/stopwatch.sim/sim_1/behav/xsim'
xvhdl --incr --relax -prj tb_stop_watch_vhdl.prj
INFO: [VRFC 10-163] Analyzing VHDL file "/home/user/Workspace/vivado/stopwatch/stopwatch.srcs/sources_1/new/Stop_watch.vhd" into library xil_defaultlib
INFO: [VRFC 10-3107] analyzing entity 'Stop_watch'
INFO: [VRFC 10-163] Analyzing VHDL file "/home/user/Workspace/vivado/stopwatch/stopwatch.srcs/sources_1/new/debounce_logic.vhd" into library xil_defaultlib
INFO: [VRFC 10-3107] analyzing entity 'debounce_logic'
INFO: [VRFC 10-163] Analyzing VHDL file "/home/user/Workspace/vivado/stopwatch/stopwatch.srcs/sim_1/new/tb_stop_watch.vhd" into library xil_defaultlib
INFO: [VRFC 10-3107] analyzing entity 'tb_stop_watch'
Waiting for jobs to finish...
No pending jobs, compilation finished.
INFO: [USF-XSim-69] 'compile' step finished in '1' seconds
INFO: [USF-XSim-3] XSim::Elaborate design
INFO: [USF-XSim-61] Executing 'ELABORATE' step in '/home/user/Workspace/vivado/stopwatch/stopwatch.sim/sim_1/behav/xsim'
xelab --incr --debug typical --relax --mt 8 -L xil_defaultlib -L secureip --snapshot tb_stop_watch_behav xil_defaultlib.tb_stop_watch -log elaborate.log
Vivado Simulator v2025.2
Copyright 1986-2022 Xilinx, Inc. All Rights Reserved.
Copyright 2022-2025 Advanced Micro Devices, Inc. All Rights Reserved.
Running: /home/user/Vivado/2025.2/Vivado/bin/unwrapped/lnx64.o/xelab --incr --debug typical --relax --mt 8 -L xil_defaultlib -L secureip --snapshot tb_stop_watch_behav xil_defaultlib.tb_stop_watch -log elaborate.log
Using 8 slave threads.
Starting static elaboration
Completed static elaboration
Starting simulation data flow analysis
Completed simulation data flow analysis
Time Resolution for simulation is 1ps
Compiling package std.standard
Compiling package std.textio
Compiling package ieee.std_logic_1164
Compiling package ieee.numeric_std
Compiling architecture behavioral of entity xil_defaultlib.debounce_logic
[\\debounce_logic(clk_freq=100,deb...]
Compiling architecture behavioral of entity xil_defaultlib.Stop_watch
[\\Stop_watch(clk_freq_hz=100,debo...]
Compiling architecture behavioral of entity xil_defaultlib.tb_stop_watch
Built simulation snapshot tb_stop_watch_behav
INFO: [USF-XSim-69] 'elaborate' step finished in '1' seconds
INFO: [USF-XSim-4] XSim::Simulate design
INFO: [USF-XSim-61] Executing 'SIMULATE' step in '/home/user/Workspace/vivado/stopwatch/stopwatch.sim/sim_1/behav/xsim'
INFO: [USF-XSim-98] *** Running xsim
 with args "tb_stop_watch_behav -key {Behavioral:sim_1:Functional:tb_stop_watch}
-tclbatch {tb_stop_watch.tcl} -log {simulate.log}"
INFO: [USF-XSim-8] Loading simulator feature
Time resolution is 1 ps
source tb_stop_watch.tcl

```

# set curr_wave [current_wave_config]
# if { [string length $curr_wave] == 0 } {
#   if { [llength [get_objects]] > 0 } {
#     add_wave /
#     set_property needs_save false [current_wave_config]
#   } else {
#     send_msg_id Add_Wave-1 WARNING "No top level signals found. Simulator will
start without a wave window. If you want to open a wave window go to 'File->New
Waveform Configuration' or type 'create_wave_config' in the TCL console."
#   }
# }
# run 1000ns
Note: ===== TEST START =====
Time: 0 ps Iteration: 0 Process: /tb_stop_watch/stim_proc File: /home/user/Workspace/
vivado/stopwatch/stopwatch.srcs/sim_1/new/tb_stop_watch.vhd
Note: Test 1: Press START
Time: 300 ns Iteration: 0 Process: /tb_stop_watch/stim_proc File: /home/user/
Workspace/vivado/stopwatch/stopwatch.srcs/sim_1/new/tb_stop_watch.vhd
INFO: [USF-XSim-96] XSim completed. Design snapshot 'tb_stop_watch_behav' loaded.
INFO: [USF-XSim-97] XSim simulation ran for 1000ns
launch_simulation: Time (s): cpu = 00:00:07 ; elapsed = 00:00:05 . Memory (MB): peak =
10264.461 ; gain = 72.121 ; free physical = 5586 ; free virtual = 30967
run 10 us
Note: Test 2: Press STOP
Time: 5500 ns Iteration: 0 Process: /tb_stop_watch/stim_proc File: /home/user/
Workspace/vivado/stopwatch/stopwatch.srcs/sim_1/new/tb_stop_watch.vhd
Note: Test 3: Press START (Resume)
Time: 7700 ns Iteration: 0 Process: /tb_stop_watch/stim_proc File: /home/user/
Workspace/vivado/stopwatch/stopwatch.srcs/sim_1/new/tb_stop_watch.vhd
Note: ===== TEST FINISHED =====
Time: 10900 ns Iteration: 0 Process: /tb_stop_watch/stim_proc File: /home/user/
Workspace/vivado/stopwatch/stopwatch.srcs/sim_1/new/tb_stop_watch.vhd
run 10 us

```

Implementation Results:

The synthesized design was implemented on the target FPGA. The implementation results confirm successful placement and routing of the design while meeting timing constraints

*** Running vivado

with args -log Stop_watch.vds -m64 -product Vivado -mode batch -messageDb
vivado.pb -notrace -source Stop_watch.tcl

***** Vivado v2025.2 (64-bit)

**** SW Build 6299465 on Fri Nov 14 12:34:56 MST 2025

**** IP Build 6300035 on Fri Nov 14 10:48:45 MST 2025

**** SharedData Build 6298862 on Thu Nov 13 04:50:51 MST 2025

**** Start of session at: Fri Jan 30 15:49:44 2026

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source Stop_watch.tcl -notrace

Command: read_checkpoint -auto_incremental -incremental /home/devanshu/
Workspace/vivado/stopwatch/stopwatch.srscs/utils_1/imports/synth_1/Stop_watch.dcp
INFO: [Vivado 12-5825] Read reference checkpoint from /home/devanshu/Workspace/
vivado/stopwatch/stopwatch.srscs/utils_1/imports/synth_1/Stop_watch.dcp for
incremental synthesis

INFO: [Vivado 12-7989] Please ensure there are no constraint changes

Command: synth_design -top Stop_watch -part xc7z010iclg225-1L

Starting synth_design

Attempting to get a license for feature 'Synthesis' and/or device 'xc7z010i'

INFO: [Common 17-349] Got license for feature 'Synthesis' and/or device 'xc7z010i'

INFO: [Designutils 20-5440] No compile time benefit to using incremental synthesis; A full
resynthesis will be run

INFO: [Designutils 20-4379] Flow is switching to default flow due to incremental criteria
not met. If you would like to alter this behaviour and have the flow terminate instead,
please set the following parameter config_implementation {autoIncr.Synth.RejectBehavior
Terminate}

INFO: [Synth 8-7079] Multithreading enabled for synth_design using a maximum of 7
processes.

INFO: [Synth 8-7078] Launching helper process for spawning children vivado processes

INFO: [Synth 8-7075] Helper process launched with PID 8218

Starting Synthesize : Time (s): cpu = 00:00:02 ; elapsed = 00:00:03 . Memory (MB): peak =
2036.418 ; gain = 467.672 ; free physical = 4456 ; free virtual = 29874

INFO: [Synth 8-638] synthesizing module 'Stop_watch' [/home/devanshu/Workspace/
vivado/stopwatch/stopwatch.srscs/sources_1/new/Stop_watch.vhd:21]

Parameter CLK_FREQ bound to: 100000000 - type: integer

Parameter DEBOUNCE_MS bound to: 20 - type: integer

INFO: [Synth 8-3491] module 'debounce_logic' declared at '/home/devanshu/Workspace/
vivado/stopwatch/stopwatch.srscs/sources_1/new/debounce_logic.vhd:7' bound to
instance 'Debounce_Start' of component 'debounce_logic' [/home/devanshu/Workspace/
vivado/stopwatch/stopwatch.srscs/sources_1/new/Stop_watch.vhd:48]

INFO: [Synth 8-638] synthesizing module 'debounce_logic' [/home/devanshu/Workspace/vivado/stopwatch/stopwatch.srcs/sources_1/new/debounce_logic.vhd:19]

Parameter CLK_FREQ bound to: 100000000 - type: integer

Parameter DEBOUNCE_MS bound to: 20 - type: integer

INFO: [Synth 8-256] done synthesizing module 'debounce_logic' (0#1) [/home/devanshu/Workspace/vivado/stopwatch/stopwatch.srcs/sources_1/new/debounce_logic.vhd:19]

Parameter CLK_FREQ bound to: 100000000 - type: integer

Parameter DEBOUNCE_MS bound to: 20 - type: integer

INFO: [Synth 8-3491] module 'debounce_logic' declared at '/home/devanshu/Workspace/vivado/stopwatch/stopwatch.srcs/sources_1/new/debounce_logic.vhd:7' bound to instance 'Debounce_Stop' of component 'debounce_logic' [/home/devanshu/Workspace/vivado/stopwatch/stopwatch.srcs/sources_1/new/Stop_watch.vhd:52]

Parameter CLK_FREQ bound to: 100000000 - type: integer

Parameter DEBOUNCE_MS bound to: 20 - type: integer

INFO: [Synth 8-3491] module 'debounce_logic' declared at '/home/devanshu/Workspace/vivado/stopwatch/stopwatch.srcs/sources_1/new/debounce_logic.vhd:7' bound to instance 'Debounce_Reset' of component 'debounce_logic' [/home/devanshu/Workspace/vivado/stopwatch/stopwatch.srcs/sources_1/new/Stop_watch.vhd:56]

INFO: [Synth 8-256] done synthesizing module 'Stop_watch' (0#1) [/home/devanshu/Workspace/vivado/stopwatch/stopwatch.srcs/sources_1/new/Stop_watch.vhd:21]

Finished Synthesize : Time (s): cpu = 00:00:03 ; elapsed = 00:00:04 . Memory (MB): peak = 2121.387 ; gain = 552.641 ; free physical = 4359 ; free virtual = 29780

Finished Constraint Validation : Time (s): cpu = 00:00:04 ; elapsed = 00:00:04 . Memory (MB): peak = 2133.262 ; gain = 564.516 ; free physical = 4355 ; free virtual = 29776

Start Loading Part and Timing Information

Loading part: xc7z010iclg225-1L

INFO: [Device 21-403] Loading part xc7z010iclg225-1L

Finished Loading Part and Timing Information : Time (s): cpu = 00:00:04 ; elapsed = 00:00:04 . Memory (MB): peak = 2133.262 ; gain = 564.516 ; free physical = 4355 ; free virtual = 29776

INFO: [Synth 8-802] inferred FSM for state register 'current_state_reg' in module 'Stop_watch'

State	New Encoding	Previous Encoding
idle	001	00
running	010	01
stopped	100	10

INFO: [Synth 8-3354] encoded FSM with state register 'current_state_reg' using encoding 'one-hot' in module 'Stop_watch'

Finished RTL Optimization Phase 2 : Time (s): cpu = 00:00:04 ; elapsed = 00:00:04 .
Memory (MB): peak = 2150.176 ; gain = 581.430 ; free physical = 4332 ; free virtual = 29756

No constraint files found.

Start RTL Component Statistics

Detailed RTL Component Info :

+---Adders :

2 Input	27 Bit	Adders := 1
2 Input	8 Bit	Adders := 1

+---Registers :

27 Bit	Registers := 1
8 Bit	Registers := 1
1 Bit	Registers := 13

+---Muxes :

3 Input	3 Bit	Muxes := 1
2 Input	1 Bit	Muxes := 3
3 Input	1 Bit	Muxes := 1

Finished RTL Component Statistics

Start Part Resource Summary

Part Resources:

DSPs: 80 (col length:40)

BRAMs: 120 (col length: RAMB18 40 RAMB36 20)

Finished Part Resource Summary

No constraint files found.

Start Cross Boundary and Area Optimization

Finished Cross Boundary and Area Optimization : Time (s): cpu = 00:00:06 ; elapsed = 00:00:06 . Memory (MB): peak = 2289.184 ; gain = 720.438 ; free physical = 4139 ; free virtual = 29601

No constraint files found.

Start Timing Optimization

Finished Timing Optimization : Time (s): cpu = 00:00:06 ; elapsed = 00:00:06 . Memory (MB): peak = 2289.184 ; gain = 720.438 ; free physical = 4132 ; free virtual = 29593

Start Technology Mapping

Finished Technology Mapping : Time (s): cpu = 00:00:06 ; elapsed = 00:00:06 . Memory (MB): peak = 2289.184 ; gain = 720.438 ; free physical = 4132 ; free virtual = 29593

Start IO Insertion

Start Flattening Before IO Insertion

Finished Flattening Before IO Insertion

Start Final Netlist Cleanup

Finished Final Netlist Cleanup

Finished IO Insertion : Time (s): cpu = 00:00:08 ; elapsed = 00:00:08 . Memory (MB): peak = 2468.996 ; gain = 900.250 ; free physical = 4002 ; free virtual = 29464

Start Renaming Generated Instances

Finished Renaming Generated Instances : Time (s): cpu = 00:00:08 ; elapsed = 00:00:08 . Memory (MB): peak = 2468.996 ; gain = 900.250 ; free physical = 4002 ; free virtual = 29464

Start Rebuilding User Hierarchy

Finished Rebuilding User Hierarchy : Time (s): cpu = 00:00:08 ; elapsed = 00:00:08 . Memory (MB): peak = 2468.996 ; gain = 900.250 ; free physical = 4002 ; free virtual = 29464

Start Renaming Generated Ports

Finished Renaming Generated Ports : Time (s): cpu = 00:00:08 ; elapsed = 00:00:08 . Memory (MB): peak = 2468.996 ; gain = 900.250 ; free physical = 4002 ; free virtual = 29464

Start Handling Custom Attributes

Finished Handling Custom Attributes : Time (s): cpu = 00:00:08 ; elapsed = 00:00:08 .
Memory (MB): peak = 2468.996 ; gain = 900.250 ; free physical = 4002 ; free virtual = 29464

Start Renaming Generated Nets

Finished Renaming Generated Nets : Time (s): cpu = 00:00:08 ; elapsed = 00:00:08 .
Memory (MB): peak = 2468.996 ; gain = 900.250 ; free physical = 4002 ; free virtual = 29464

Start Writing Synthesis Report

Report BlackBoxes:

BlackBox name	Instances

Report Cell Usage:

	Cell	Count
1	BUFG	1
2	CARRY4	31
3	LUT1	17
4	LUT2	32
5	LUT3	9
6	LUT4	2
7	LUT5	3
8	LUT6	8
9	FDRE	114
10	IBUF	4
11	OBUF	8

Report Instance Areas:

	Instance	Module	Cells
1	top		229
2	Debounce_Reset	debounce_logic	53
3	Debounce_Start	debounce_logic_0	50
4	Debounce_Stop	debounce_logic_1	50

Finished Writing Synthesis Report : Time (s): cpu = 00:00:08 ; elapsed = 00:00:08 .
Memory (MB): peak = 2468.996 ; gain = 900.250 ; free physical = 4002 ; free virtual = 29464

Synthesis finished with 0 errors, 0 critical warnings and 0 warnings.

Synthesis Optimization Runtime : Time (s): cpu = 00:00:08 ; elapsed = 00:00:08 . Memory (MB): peak = 2468.996 ; gain = 900.250 ; free physical = 4002 ; free virtual = 29464

Synthesis Optimization Complete : Time (s): cpu = 00:00:08 ; elapsed = 00:00:08 .

Memory (MB): peak = 2469.004 ; gain = 900.250 ; free physical = 4002 ; free virtual = 29464

INFO: [Project 1-571] Translating synthesized netlist

Netlist sorting complete. Time (s): cpu = 00:00:02 ; elapsed = 00:00:00.01 . Memory (MB): peak = 2482.871 ; gain = 0.000 ; free physical = 4086 ; free virtual = 29550

INFO: [Netlist 29-17] Analyzing 31 Unisim elements for replacement

INFO: [Netlist 29-28] Unisim Transformation completed in 0 CPU seconds

INFO: [Project 1-570] Preparing netlist for logic optimization

INFO: [Opt 31-138] Pushed 0 inverter(s) to 0 load pin(s).

Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed = 00:00:00 . Memory (MB): peak = 2621.547 ; gain = 0.000 ; free physical = 4021 ; free virtual = 29496

INFO: [Project 1-111] Unisim Transformation Summary:

No Unisim elements were transformed.

Synth Design complete | Checksum: f5a5cecb

INFO: [Common 17-83] Releasing license: Synthesis

25 Infos, 0 Warnings, 0 Critical Warnings and 0 Errors encountered.

synth_design completed successfully

synth_design: Time (s): cpu = 00:00:11 ; elapsed = 00:00:10 . Memory (MB): peak = 2621.582 ; gain = 1052.836 ; free physical = 4009 ; free virtual = 29484

INFO: [Common 17-2834] synth_design peak Physical Memory [PSS] (MB): overall = 1783.484; main = 1783.484; forked = 0.000

INFO: [Common 17-2834] synth_design peak Virtual Memory [VSS] (MB): overall = 2621.551; main = 2621.551; forked = 0.000

Write ShapeDB Complete: Time (s): cpu = 00:00:00.01 ; elapsed = 00:00:00 . Memory (MB): peak = 2645.559 ; gain = 0.000 ; free physical = 4009 ; free virtual = 29484

INFO: [Common 17-1381] The checkpoint '/home/devanshu/Workspace/vivado/stopwatch/stopwatch.runs/synth_1/Stop_watch.dcp' has been generated.

INFO: [Vivado 12-24828] Executing command : report_utilization -file

Stop_watch_utilization_synth.rpt -pb Stop_watch_utilization_synth.pb

INFO: [Common 17-206] Exiting Vivado at Fri Jan 30 15:49:56 2026...

Here is a snapshot of the final simulation and implementation report -

Report	Type	Options	Modified	Size
Synthesis				
Synth Design (synth_design)				
Utilization - Synth Design	report_utilization		1/30/26, 3:49 PM	8.1 KB
synthesis_report			1/30/26, 3:49 PM	17.6 KB
Implementation				
impl_1				
Design Initialization (init_design)				
Timing Summary - Design Initialization	report_timing_summary	max_paths = 10; report_unconstrained = true;		
Opt Design (opt_design)				
DRC - Opt Design	report_drc		1/30/26, 3:50 PM	3.7 KB
Timing Summary - Opt Design	report_timing_summary	max_paths = 10; report_unconstrained = true;		
Power Opt Design (power_opt_design)				
Timing Summary - Power Opt Design	report_timing_summary	max_paths = 10; report_unconstrained = true;		
Place Design (place_design)				
IO - Place Design	report_io		1/30/26, 3:50 PM	66.9 KB
Utilization - Place Design	report_utilization		1/30/26, 3:50 PM	10.4 KB
Control Sets - Place Design	report_control_sets	verbose = true;	1/30/26, 3:50 PM	4.5 KB
Incremental Reuse - Place Design	report_incremental_reuse			
Incremental Reuse - Place Design	report_incremental_reuse			
Timing Summary - Place Design	report_timing_summary	max_paths = 10; report_unconstrained = true;		
Post-Place Power Opt Design (post_place_power_opt_design)				
Timing Summary - Post-Place Power Opt Design	report_timing_summary	max_paths = 10; report_unconstrained = true;		
Post-Place Phys Opt Design (phys_opt_design)				
Timing Summary - Post-Place Phys Opt Design	report_timing_summary	max_paths = 10; report_unconstrained = true;		
Route Design (route_design)				
DRC - Route Design	report_drc		1/30/26, 3:50 PM	3.7 KB
Methodology - Route Design	report_methodology		1/30/26, 3:50 PM	19.8 KB
Power - Route Design	report_power		1/30/26, 3:50 PM	8.9 KB
Route Status - Route Design	report_route_status		1/30/26, 3:50 PM	0.6 KB
Timing Summary - Route Design	report_timing_summary	max_paths = 10; routable_nets = true; report_unconstrained = true;	1/30/26, 3:50 PM	35.1 KB
Incremental Reuse - Route Design	report_incremental_reuse			
Clock Utilization - Route Design	report_clock_utilization		1/30/26, 3:50 PM	10.5 KB
Bus Skew - Route Design	report_bus_skew	warn_on_violation = true;	1/30/26, 3:50 PM	0.9 KB
implementation_log			1/30/26, 3:50 PM	36.2 KB